

Joy Zhang

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EDUCATION

Cornell University, College of Engineering

Master of Engineering, Computer Science, GPA 3.9

May 2021

Bachelor of Science, Computer Science Major, Game Design Minor, GPA 3.5

Dec 2020

PROFESSIONAL EXPERIENCE

Hometopia Inc.

Senior Game Developer

Dec 2024 - Feb 2025

Junior Game Developer

Jul 2024 - Dec 2024

- Contributed to the development of a multiplayer house-building PC sandbox game in **Unity** by writing clean, well-organized, and easily maintainable **C#** code
- Took ownership of UI development, collaborated with designers to give game a cohesive look and feel
- Lead and mentored a small team of junior developers, delegated tasks, and performed code reviews
- Developed an audio system to centralize control of music, ambience, and sound effects
- Created a system for the first-time user experience and tutorial
- Improved multiplayer features, including friend list, parties, chat, and save file transfer and sharing
- Investigated and fixed critical gameplay, multiplayer, and UI bugs, and developed new features
- Took responsibility for developing demo of a separate virtual pet mobile game and setting up structure of game by implementing a state machine for main game loop and connecting different game systems

Northrop Grumman

ODIN, Game Developer

Jan 2022 - Jun 2024

- Developed Omniverse apps in **Python** with Universal Scene Description to build a virtual warfighting environment by controlling drones and sensors, streamlining data generation, and visualizing data
- Built physically accurate flight simulators for Volansi and Spirit drones with Omniverse's integrated **PhysX** physics engine and performed manual 3D vector math calculations
- Integrated Omniverse simulation with external AI code and live drones running in real-time
- Created a data generation pipeline for Omniverse to train AI/ML models by developing a suite of tools in **Python** to create, control, and randomize scenes, assets, and perceptual variables

COAGEN, Software Engineer

Aug 2021 - Dec 2021

- Built demo in **Java** for modified batch scheduling algorithm accounting for interruptions predicted by a long short-term memory (LSTM) neural network for long-term resilience against adhoc task disruptions

PROJECTS

Game Design Initiative at Cornell

Jan 2019 - May 2021

- Developed gameplay mechanics, physics, and programmatic animations with the Cornell University Games Library in **C++** and **libGDX** in **Java** as a gameplay programmer to make games in teams of 7
- Accepted to BostonFIG Fest in 2019 with computer game *Starstruck*, a cooperative puzzle platformer set in space, and in 2020 with mobile game *Spectacle!*, a one-tap pinball-inspired puzzle game
- Guided students through entire process of designing and creating digital games as a Teaching Assistant

Global Summit

Jun 2020 - Aug 2020

- Created **Pygame** simulation for an unmanned wildfire identification and suppression system

Cornell Big Red Hacks

Sep 2019

- Awarded Best Design Hack for *One Call Away*, a short story platformer that highlights the importance of mental health created in **Unity** in 36 hours in a team of 4

SKILLS

Competencies: Agile, Algorithms, Data structures, Game development, Gameplay programming, Object-oriented programming, Physics simulation, UI programming

Programming Languages: C, C#, C++, Java, MATLAB, OCaml, Perl, PHP, Python

Specialties: Blender, Box2D, Construct3, libGDX, Nvidia Omniverse, PhysX, Pygame, Unity

Other Software: Adobe (Character Animator, Illustrator), ClickUp, CSS, Docker, Eclipse, Git, HTML, IntelliJ IDEA, Jira, LaTeX, Linux, Maven, Plastic SCM, Robot Framework, Visual Studio, Visual Studio Code